Roll No. | | | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 09

B.Tech. (CE) (Sem.–7)

SUSTAINABLE ENGINEERING AND TECHNOLOGIES

Subject Code : PECE-702F-18

M.Code : 90628

Date of Examination : 23-12-2022

Time : 3 Hrs. Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.
3. SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.

SECTION-A

1. Write briefly :

(a) What is meant by integrated life cycle design of materials?

(b) What is BREEAM?

(c) Enumerate the major indentation of Air act.

(d) Explain the term Industrial Ecology.

(e) What do you mean by climate change?

(f) Explain the significance of carbon footprints.

(g) Suggest any three sustainable methods for waste water management system.

(h) How can be energy derived from oceans?

(i) Explain zero waste concept.

(j) What Green Engineering stands for?

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SECTION-B

2. Explain contribution of buildings towards global warming.

3. Give the characteristics of any three waste materials that can be used as a substitute of cement.

4. Define Sustainability. What is the need of sustainable development?

5. *'Industrialization was the major turning point in the social life'*. How did industrialization change the way of human life?

6. Write a note on embodied energy and embodied carbon-di-oxide emission.

SECTION-C

7. Explain the mandatory rule in GRIHA related to construction.

8. (a) Explain the various challenges for the sustainable development.

   (b) What is meant by carbon sequestration?

9. Why renewable energy is important? Explain **any five** renewable energy sources.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.

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